

Targeted therapies and immunotherapy for stomach cancer

Targeted therapies interfere with the way cells grow and divide. Immunotherapy uses the immune system to recognise and kill cancer cells.

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What are targeted therapies?

Targeted therapy uses drugs to find and attack cancer cells. There are many different types of [targeted therapy](#) drugs. Each type targets something in or around the cancer cell that is helping it grow and survive.

Targeted therapy drugs may be used to treat advanced [stomach cancer](#).

Targeted therapy drugs for stomach cancer

Sometimes a targeted therapy drug called [trastuzumab](#) is given with [chemotherapy](#) to treat advanced stomach cancer. Targeted therapy drugs target something in or around the cancer cell that is helping it grow and survive.

Trastuzumab only works for people with a stomach cancer that has high levels of a protein called HER2. 10 to 15 out of every 100 people with stomach cancer (10 to 15%) have a HER2-positive cancer. Tests can be done on tissue taken at a biopsy or during surgery to find out if the cancer cells have high levels of HER2.

Trastuzumab attaches to the HER2 proteins on the surface of the cancer cells. This stops the cancer cells from dividing and growing.

How trastuzumab is given

Trastuzumab is given as a drip (infusion) every 3 weeks. If it works well, your cancer doctor may decide to keep giving it after your chemotherapy finishes. The side effects of trastuzumab are usually mild.

What is immunotherapy?

Immunotherapy drugs use the immune system to find and attack cancer cells.

You may be offered an immunotherapy drug called nivolumab. Nivolumab targets and blocks a protein (receptor) called PD-1 on the surface of T cells, which are part of the immune system. It also blocks another protein called PD-L1. This helps the immune system to find and attack the cancer cells.

Some people have tests on the cancer cells to check the levels of PD-L1. The results can tell your doctor how likely this type of immunotherapy drug is to be helpful. Immunotherapy drugs can still work for you even if results do not show high levels of PD-L1.

You may be given nivolumab along with chemotherapy if you have advanced stomach cancer. Your cancer doctor can tell you more about whether nivolumab might be helpful in your situation.

Some people with advanced cancer of the stomach may be offered another immunotherapy drug called pembrolizumab with chemotherapy. This will depend on the treatments you've had before and the levels of PD-L1 that are found on the cancer cells. Your cancer doctor can tell you more about whether this might be suitable for you.

You have pembrolizumab, along with chemotherapy, as a drip (infusion) into a vein. A nurse will give them to you in the chemotherapy day unit.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our stomach cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

Lordick F, Carneiro S, Cascinu T, Fleitas K, Haustermans G, Piessen A, et al. Gastric cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. *Annals of Oncology*. 2022;33(10): 1005–1020. Available from www.doi.org/10.1016/j.annonc.2022.07.004 [accessed July 2023].

National Institute for Care and Health Excellence. Oesophago-gastric cancer: assessment and management in adults NICE guideline [NG83]. 2018. Available from www.nice.org.uk/guidance/ng83 [accessed July 2023].

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help